

Linux System Performance Review

System: ITC00E0C52A6A64 Kernel: Linux 6.18.6 Collection: July 30, 2026, 4:46:13 PM-4:49:34 PM

Duration: 200 seconds CPU: 4 logical processors Memory: 3.2 GiB RAM Storage: 7.3 GB
eMMC (/dev/mmcblk0) Network: wlan0

Executive Summary

The system experienced severe memory pressure during the test. Microsoft Edge processes expanded rapidly, consumed most physical memory, and filled almost the entire 1.9 GiB compressed swap device. The resulting memory reclamation and swap activity drove elevated kernel CPU usage, I/O wait, and bursts of eMMC saturation.

The underlying CPU, network, and storage do not show clear evidence of hardware failure. The primary issue is that 3.2 GiB of RAM is insufficient for the observed Edge workload. Once browser renderer processes grew, the system entered a memory-reclamation and swap-thrashing cycle.

Summary Table

Area	Status	Key findings	Assessment
Physical memory	CRITICAL	3.2 GiB installed; available memory fell from 1.4 GiB to 317 MiB	RAM capacity is too low for the workload
Swap	CRITICAL	Usage rose from 1.2 GiB to about 1.9 GiB; only 8.3 MiB remained	Swap was effectively exhausted
CPU	WARNING / PEAK CRITICAL	Average 24.41% user, 25.16% system, 7.98% I/O wait, 42.26% idle	Load was amplified by memory and I/O work
Storage	WARNING	Average 26.5% utilization; peak 96.8%; latency spikes near 270 ms	eMMC became a secondary bottleneck
Processes	CRITICAL	Edge renderers reached about 93% CPU and 19.5% RAM individually	Edge was the immediate workload trigger
Network	HEALTHY	About 81 KiB/s receive and 28 KiB/s transmit; no RX errors or drops	Network was not the bottleneck
Overall	CRITICAL UNDER LOAD	Memory exhaustion caused swapping, storage pressure, and high system CPU	Workload exceeds practical system capacity

Detailed Analysis

1. Memory

At the beginning of collection, the system already had about 1.2 GiB of its 1.9 GiB swap in use. This means the endpoint did not begin the test from a clean, unconstrained memory state.

Beginning: RAM 3.2 GiB total, 1.8 GiB used, 1.4 GiB available; swap 1.9 GiB total, 1.2 GiB used.
End: RAM 3.2 GiB total, 2.9 GiB used, 317 MiB available; swap 1.9 GiB total, about 1.9 GiB used, 8.3 MiB free.

During only 200 seconds, available RAM declined by about 1.1 GiB, swap usage increased by roughly 700 MiB, and swap exceeded 99% utilization. Repeated simultaneous swap-in and swap-out activity indicates memory thrashing: pages were evicted and then needed again shortly afterward.

Memory conclusion: Physical RAM is the primary limitation. A 3.2 GiB endpoint has little margin for several complex tabs, Microsoft 365 web apps, browser extensions, conferencing, or multiple Chromium-based applications.

2. CPU

Across the collection, CPU averaged 24.41% user, 25.16% system, 7.98% I/O wait, and 42.26% idle. The unusually high system CPU component is consistent with memory reclamation, page faults, compressed swap processing, filesystem reads, and heavy process scheduling.

Near the end of the test, idle CPU dropped to about 14-21%. One observed core spent approximately 78% in system CPU with only 5% idle. The runnable-process count reached about 18 on a four-CPU system, indicating periods of scheduling pressure.

CPU conclusion: The processor is not necessarily undersized for normal endpoint use. Much of the observed CPU pressure was created by the memory shortage; reducing browser memory or increasing RAM should also lower CPU usage.

3. Microsoft Edge Process Activity

Process snapshot	Peak CPU	Peak memory
Edge renderer PID 75613	92.7%	14.5%
Same renderer, later snapshot	70.3%	19.5%
Edge renderer PID 75372	89.5%	7.7%
Edge renderer PID 75185	78.8%	6.8%
Main Edge process	64.3%	9.0%

A single process at 19.5% of 3.2 GiB represents roughly 640 MiB. Several browser processes were active simultaneously, so total Edge memory was substantially larger. The immediate trigger was therefore the Edge workload, while insufficient RAM was the underlying capacity constraint.

4. Storage and Disk I/O

The primary device is a 7.3 GB eMMC disk, /dev/mmcblk0. During active samples it averaged about 520 reads/s, 13.1 MiB/s read bandwidth, 38 writes/s, 2.8 MiB/s write bandwidth, 4.52 ms read latency, 11.93 ms write latency, 26.5% utilization, and queue depth near 1.84.

Peak observations included approximately 75.7 MiB/s reads, 96.8% device utilization, and latency spikes near 270-277 ms. The eMMC device became a secondary bottleneck because the memory subsystem could not retain the active working set.

5. Network and TCP

The active interface was wlan0 at 10.0.0.18 with default gateway 10.0.0.1. Traffic averaged about 81.38 KiB/s receive and 27.79 KiB/s transmit. No receive errors or receive drops were recorded, and only seven cumulative transmit drops appeared. Network bandwidth was not a material bottleneck.

Some CLOSE-WAIT and occasional SYN-SENT sockets appeared. A small temporary count is normal; a continuously rising count could indicate that an application is not releasing sockets. This is worth monitoring but is not the root cause of the performance event.

Root-Cause Assessment

1. The system began with approximately 1.2 GiB of swap already occupied.
2. Microsoft Edge expanded several renderer processes.
3. Edge consumed most available physical RAM.
4. The kernel began aggressive memory reclamation and compressed swapping.
5. Previously swapped pages were needed again, generating swap-ins.
6. Browser executable and page data were reread from eMMC.
7. eMMC utilization and latency increased.
8. Kernel/system CPU and I/O wait rose.
9. Swap reached almost 100%, leaving almost no memory safety margin.

Classification: RAM exhaustion leading to swap thrashing, with Microsoft Edge as the primary workload contributor.

Recommended Actions

Immediate

- Restart Microsoft Edge or reboot the endpoint to clear accumulated browser processes and reset swap use.
- Reduce active tabs and windows, and identify whether one particular site causes rapid memory growth.
- Disable unnecessary Edge extensions and avoid running multiple Chromium-based browsers simultaneously.
- Use Edge Task Manager (Shift+Esc) and sort by memory and CPU to identify the responsible tab, frame, or extension.

Browser and Policy Optimization

- Enable sleeping tabs and shorten the inactive-tab timeout.
- Disable background mode when Edge closes.
- Limit unnecessary startup pages and browser extensions.
- Disable preload or startup boost where appropriate.
- Restrict especially heavy sites to a controlled number of tabs.

Hardware Sizing

RAM	Assessment
4 GiB	Minimum; still constrained for modern Chromium workloads
8 GiB	Practical baseline for normal browser-heavy endpoint use
16 GiB	Preferred for conferencing, multiple applications, or heavier browser use

Additional Diagnostics for the Next Reproduction

```
cat /proc/meminfo
cat /proc/pressure/memory
cat /proc/pressure/io
cat /proc/pressure/cpu
swapon --show
```

```
zramctl  
ps -eo pid,ppid,rss,vsz,%mem,%cpu,comm,args --sort=-rss
```

To calculate total Edge resident memory:

```
ps -C msedge -o rss= | awk '{sum += $1} END {printf "Total Edge RSS: %.1f MiB\n", sum / 1024}'
```

Final Assessment

Category	Assessment
Health under light load	Acceptable
Health under tested Edge workload	Critical
Primary bottleneck	Physical memory
Secondary bottleneck	eMMC I/O caused by swapping
Primary workload contributor	Microsoft Edge renderer processes
Network bottleneck	No
Evidence of failed storage	No clear evidence
Most effective remediation	Increase RAM capacity or materially reduce the browser working set